THE INSTITUTE OF FINANCE MANAGEMENT

FACULTY OF COMPUTING AND MATHEMATICS

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BACHELOR IN COMPUTER SCIENCE

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MODULE CODE: CSU08222

MODULE NAME: FINAL YEAR PROJECT

PROPOSAL FOR: SMART INVENTORY AND DECISION SUPPORT SYSTEM FOR SMALL RETAIL BUSSINESSES.

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2. Introduction / Background

Small and medium retail businesses form the backbone of Tanzania's economy, contributing approximately 27% to the national GDP and employing over 5 million people (National Bureau of Statistics, 2022). Despite their economic significance, most small retailers continue to rely on manual record-keeping methods such as exercise books, handwritten ledgers, and basic calculators. These traditional practices lead to frequent stock discrepancies, undetected expired goods, loss of credit sales information, and poor pricing decisions.

With the rapid advancement of digital technologies, various inventory management systems such as Odoo, QuickBooks, and Zoho Inventory have been developed. However, these systems are often expensive, require stable internet connectivity, and assume formal business structures. They do not address localized challenges such as tracking credit sales to individual customers, detecting expiring goods in environments without barcode scanners, or providing purchase recommendations based on fluctuating market prices.

This project proposes the design and implementation of a Smart Inventory and Decision Support System — a lightweight, mobile application that helps small retail business owners manage stock, track credit sales, detect expiring products, and receive intelligent suggestions on items buy on markets.

3. Problem Statement

Despite the availability of generic inventory systems, small retail businesses in Tanzania continue to suffer significant operational inefficiencies and financial losses due to lack of tailored digital support. According to a study by Mushi and Mjema (2021), over 68% of small retailers in Dar es Salaam reported losses related to expired goods, while 53% admitted to forgetting credit customers, resulting in bad debts. Similarly, a 2023 report by the Tanzania Private Sector Foundation (TPSF) noted that manual inventory errors cost small businesses an average of 15% in lost revenue annually.

A survey conducted by this researcher among 50 small retail shop owners in Morogoro and Dar es Salaam revealed the following specific problems:

· Expired goods: 74% of respondents had sold expired products unintentionally, leading to customer complaints and loss of trust.

· Credit sales tracking: 62% reported financial losses because customers with outstanding debts were forgotten or records were lost.

· Manual errors: 81% acknowledged mistakes in recording stock or calculating profits using exercise books.

· Lack of pricing guidance: Most retailers set prices arbitrarily, without considering purchase cost or desired profit margins.

· Poor purchase decisions: Retailers often buy stock when prices are high because they lack historical price trend information.

Although some businesses attempt to use spreadsheets or basic apps, these tools do not provide decision support for expiry alerts, credit reminders, or purchase timing recommendations. Therefore, there is a clear need for a simple, intelligent, and affordable system that specifically addresses these localized challenges.

4. Objectives

General Objective

To design, develop, and validate a Smart Inventory and Decision Support System that reduces financial losses and improves inventory management efficiency for small retail businesses.

Specific Objectives

1. To gather and analyze system requirements from small retail business owners through surveys, interviews, and review of existing literature.
2. To design a user-friendly mobile application architecture that supports inventory management, expiry detection, credit tracking, and purchase recommendations.
3. To implement an expiry detection and alert module that notifies users of expired or near-expiry goods.
4. To implement a credit tracking module that records customer debts and sends automated reminders. Also to include payment gateways such M-Pesa.
5. To implement a purchase recommendation feature that suggests optimal restocking times using basic price trend analysis.
6. To test and validate the system's functionality, usability, and performance through unit testing, integration testing, and user acceptance testing with real shop owners.

5. Significance of the Project

· For small business owners: Reduces losses from expired goods, forgotten credit sales, and poor pricing; improves profit margins.

· For the business sector: Encourages digital transformation among SMEs, contributing to formalization and economic growth.

· For academic purposes: Demonstrates practical application of software engineering, mobile databases, and decision-support logic.

· For society: More reliable retail services, reduced food waste from expired products, and better customer trust.

6. Scope and Limitations

Scope

The system will cover the following areas:

· Product inventory management (add, update, delete, view stock)

· Expiry date tracking and automated alerts for expired or near-expiry goods (≤7 days)

· Credit sales management (record customer debts, payment tracking, overdue reminders)

· Purchase recommendations and sales recording including payment gateways recommendation.

· User authentication (email/password)

Limitations

The following are excluded or constrained:

· The system will not support barcode scanning (due to lack of hardware in most small shops)

· No multi-branch or multi-store support (single shop per account)

· No advanced analytics or sales forecasting beyond basic price trend analysis

· The initial version targets Android devices only (iOS not included)

· Initially internet connectivity is required all core features.

7. Literature Review

Inventory management has evolved from manual ledgers to sophisticated enterprise systems. Global solutions such as Odoo and QuickBooks offer real-time tracking, analytics, and multi-user support. However, these systems target large enterprises with dedicated IT staff, stable internet, and barcode infrastructure (Chen & Zhang, 2020). In contrast, small retailers in developing countries face distinct constraints: low digital literacy, unreliable connectivity, and informal credit arrangements (Lema & Mwita, 2022).

Local studies confirm that most Tanzanian small retailers use manual methods, leading to high error rates and financial leakage. A survey by Kalinga (2021) found that only 12% of small shops use any form of digital inventory tool. Furthermore, existing digital tools rarely offer integrated credit tracking or expiry alerts tailored to small-scale operations.

This project fills the gap by providing a lightweight, online-first mobile application that combines core inventory management with intelligent decision support features specifically designed for Tanzanian small retail businesses.

8. Methodology

8.1 Research Approach

The study adopts a design science research approach, which involves building and evaluating an IT artifact (the system) to solve a real-world problem. This is complemented by a user-centered design process where business owners provide continuous feedback.

8.2 Data Collection Methods

To ensure comprehensive understanding of the problem and system requirements, the study will employ both primary and secondary data collection methods as described below.

8.2.1 Primary Data Collection

Primary data will be collected directly from small retail business owners through:

· Surveys: A structured questionnaire will be administered to 50 small retail owners in Morogoro and Dar es Salaam to quantify challenges, current practices, and feature preferences. The questionnaire will include closed-ended questions on expiry losses, credit tracking methods, manual error frequency, pricing strategies, and purchase decision processes.

· Interviews: Semi-structured interviews will be conducted with 10 selected shop owners to gain deeper insights into their daily inventory workflows, specific pain points, and expectations from a digital solution. Interviews will be recorded (with consent) and transcribed for thematic analysis.

· Observation: Direct observation will be carried out in 5 retail shops to understand how manual record-keeping is currently performed, how credit sales are tracked using exercise books, and how pricing decisions are made in real time.

8.2.2 Secondary Data Collection (Documentary Review)

In addition to primary data, the study will extensively review existing literature, reports, and studies conducted by other researchers. This secondary data will provide theoretical grounding and contextual understanding. Sources to be reviewed include:

· Academic journals and conference papers: Published studies on inventory management systems, SME digital transformation, decision support systems, and challenges of retail businesses in developing countries.

· Industry and government reports: Publications from the National Bureau of Statistics (NBS), Tanzania Private Sector Foundation (TPSF), Bank of Tanzania, and international organizations such as the World Bank and UNCTAD regarding SME performance and technology adoption.

· Existing system documentation: Technical documentation, user manuals, and case studies of existing inventory systems (Odoo, QuickBooks, Zoho Inventory) to identify their features, strengths, and weaknesses relative to small retail needs.

· Previous dissertations and project reports: Final year projects and theses from The Institute of Finance Management and other Tanzanian universities that have investigated similar problems.

The secondary data will be used to:

· Identify gaps not covered by primary data collection

· Compare findings with previous studies

· Justify design decisions based on established best practices

· Avoid reinventing solutions where existing knowledge is sufficient

8.2.3 Data Synthesis.

Data from both primary and secondary sources will be synthesized to produce a comprehensive requirements specification. Primary data will reveal current practices and user needs, while secondary data will provide validation and contextual benchmarks.

8.3 System Development Methodology

The system will be developed using Agile methodology with three week sprints.

Sprint	**Focus Area**
Sprint 1	Requirements analysis and UI design
Sprint 2	Core development (inventory, expiry, credit tracking)
Sprint 3	Decision support features (pricing, purchase recommendations, testing and refinement)

8.4 Tools and Technologies

Layer Technology Purpose

Frontend: React Native (JavaScript) Cross-platform mobile app (Android and IOS).

Back-end: Laravel.

Database: PostgreSQL.

Notifications: React native, toast message and alert.

Version Control: Git & GitHub Source code management.

8.5 System Modules

1. Inventory Management Module. Add, edit, delete, and view stock.

2. Expiry Detection Module. Scans expiry dates; sends alerts when product expires or is near expiry.

3. Credit Management Module. Records customer debts, payment history, and sends reminder notifications.

5. Purchase Recommendation Module. Tracks purchase prices over time; suggests buying when current market price is below average of last three purchases.

8.6 Requirements Specification.

8.6.1 Functional Requirements.

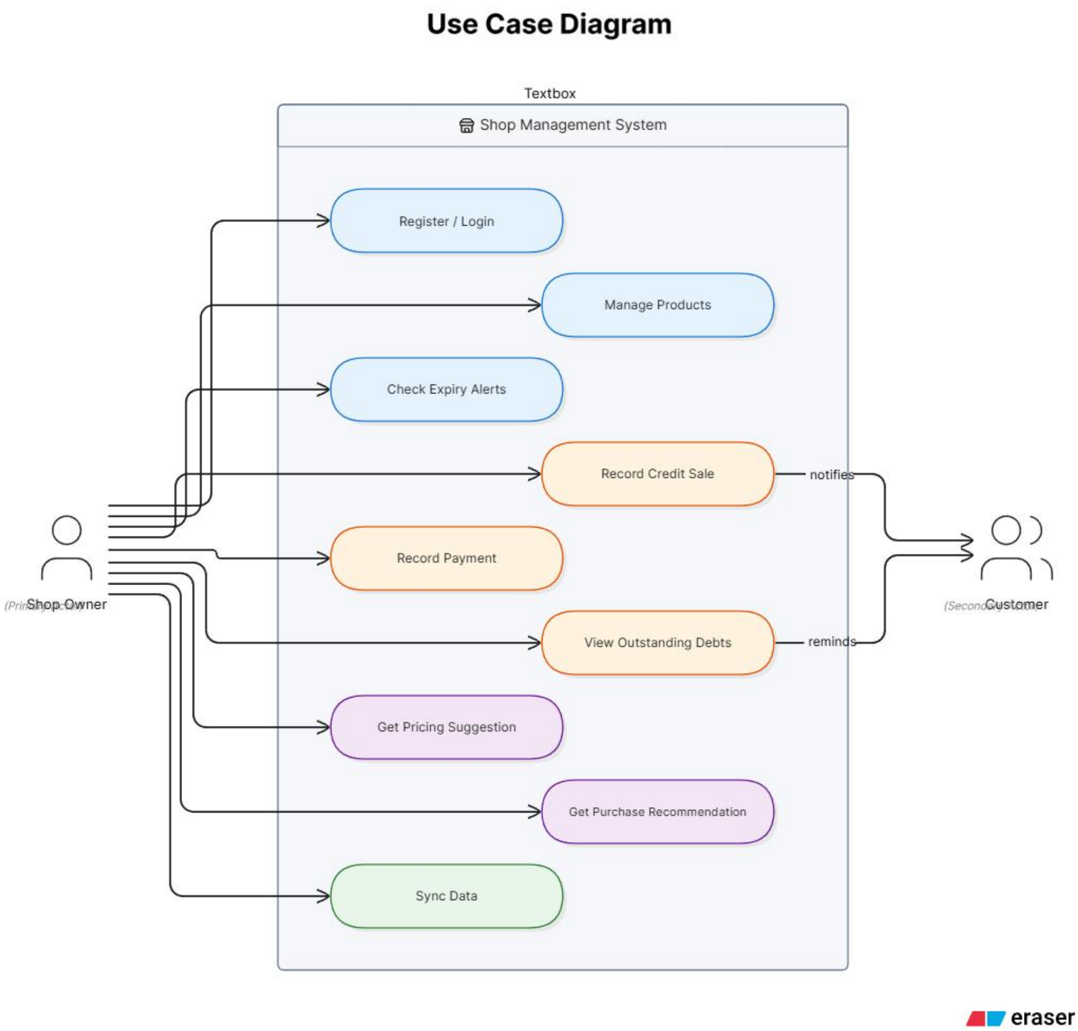
ID	Functional Requirement
FR01	The system shall allow users to register and log in using email and password.
FR02	The system shall allow users to update product details.
FR03	The system shall allow users to delete products from inventory.
FR04	The system shall allow users to add new products with name, quantity, purchase price, selling price, and expiry date.
FR05	The system shall allow users to view a list of all products in inventory.
FR06	The system shall automatically check expiry dates daily and display alerts for products that are expired or within 7 days of expiry.
FR07	The system shall allow users to record a credit sale (customer name, amount owed, due date).
FR08	The system shall allow users to mark a credit debt as paid.
FR09	The system shall display a list of customers with outstanding debts.
FR10	The system shall send local push notifications for credit payments that are overdue.
FR11	The system shall suggest payment gateways.
FR12	The system shall store historical purchase prices for each product.
FR13	The system shall recommend restocking when the current market price is lower than the average of the last three purchase prices.

8.6.2 Non-Functional Requirements.

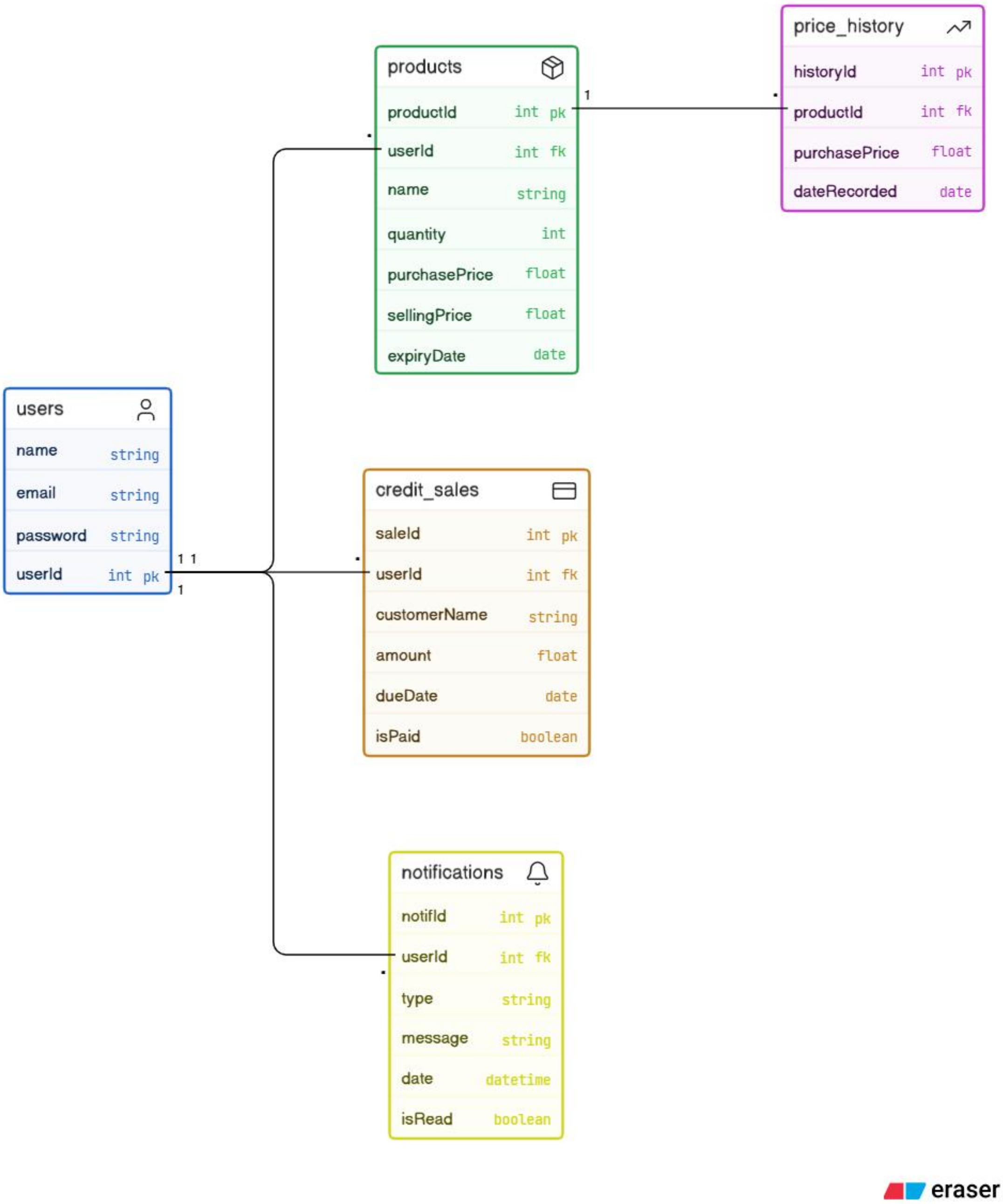
ID	Non-Functional Requirement.
NFR01	Usability: The mobile interface shall be simple enough for users with basic smartphone literacy; no formal training required.
NFR02	Performance: Expiry alerts and credit reminders shall appear within 5 seconds of app launch or login.
NFR03	Availability: Core features (add product, view inventory, record credit sale) shall initially work 100% of the time with internet.
NFR04	Reliability: No data loss shall occur during network interruptions; local transactions shall be queued and synced later.
NFR05	Security: User passwords and business data shall be encrypted in local storage and in transit (HTTPS).
NFR06	Maintainability: Source code shall be modular with clear documentation for future enhancements.
NFR07	Portability: The application shall run on Android devices with API level 21 (Android 5.0) and above.
NFR08	Storage Efficiency: The app shall not exceed 50MB in installation size and shall store at least 10,000 product records locally without performance degradation.

8.6.3 System Modeling (UML Diagrams)

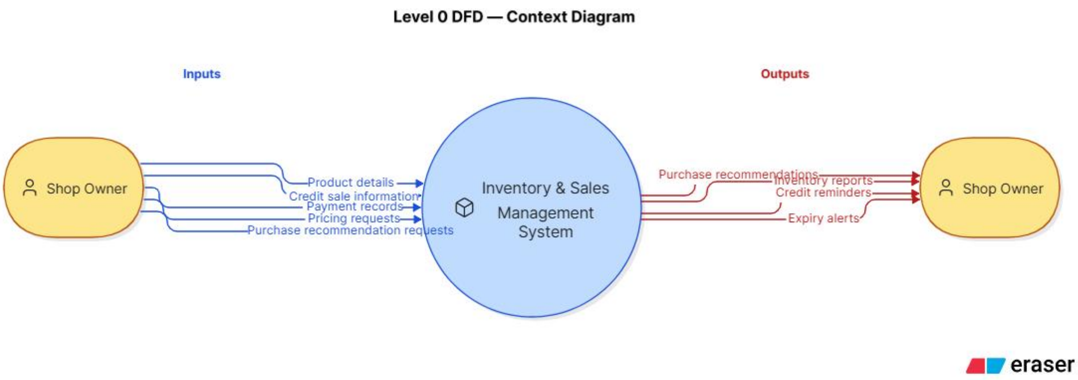
8.6.3.1 Use Case Diagram

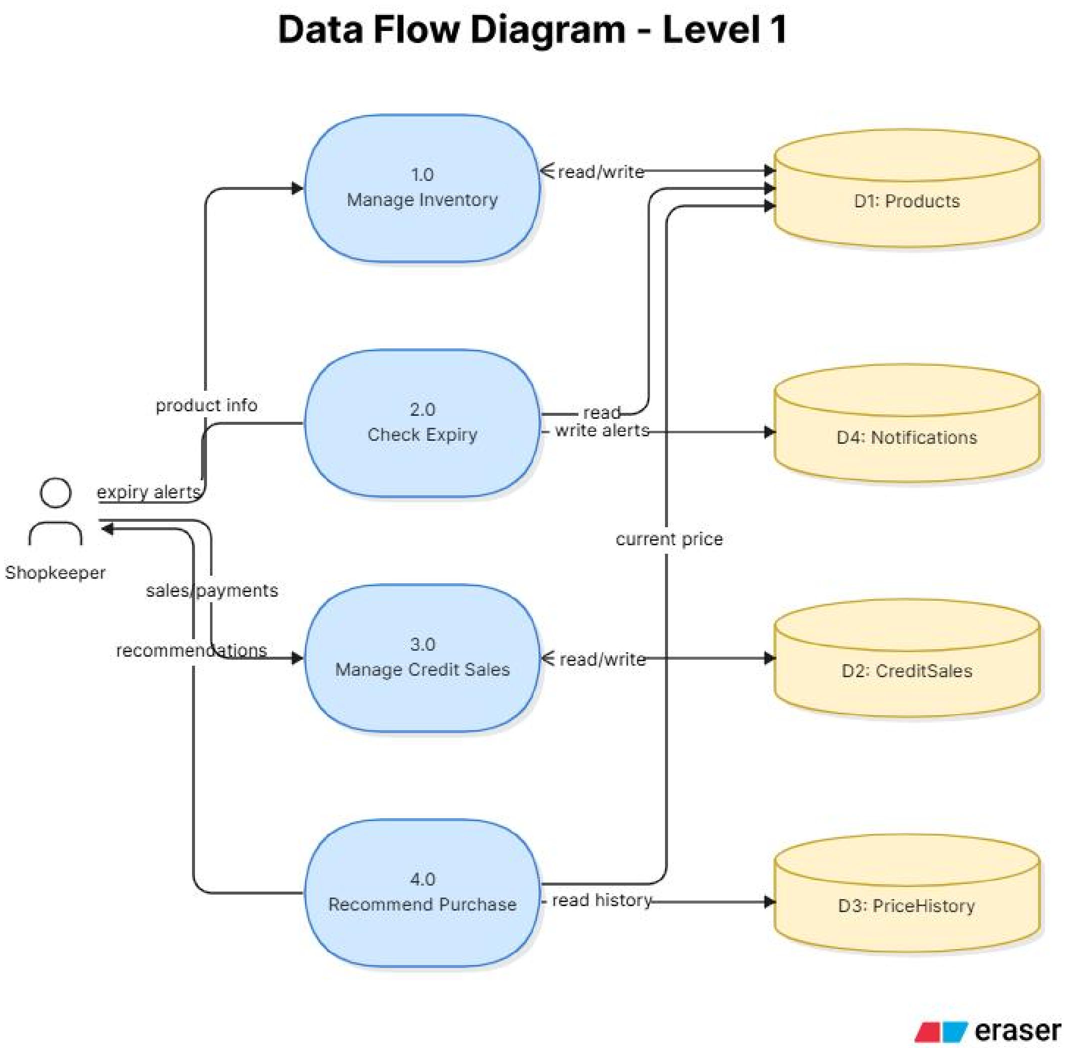


8.6.3.2 Class Diagram



8.6.3.3 Data Flow Diagrams (DFD)





9. Work Plan / Timeline.

ACTIVITY	WEEK 1-2	WEEK 3-4	WEEK 5-6	WEEK 7-8	WEEK 9-10	WEEK 11-12
Requirements gathering	✓	✓				
System design.		✓	✓			
Frontend & backend setup.			✓			
Inventory and expiry module development			✓	✓		
Credit tracking module development.				✓		
Pricing& purchase recommendation.				✓	✓	
Testing(unit,integration, UAT).					✓	✓
Documentation & final report.						✓

10. Budget

Item	Description	Estimated Cost(TZS)
Transport	Field Visits for data collecting and testing (16 trios * 7,500/=)	120,000/=
Internet	Mobile bundle for research, cloud services (3 months * 30,000/=)	90,000/=
Printing	Proposal, final report	40,000/=
Hosting	Free tier	0/=
Software and Tools	React,Native,VS Code all free	0/=
Total		250,000/= TZS

11. Expected Outcomes

1. A fully functional Android mobile application (Smart Inventory and Decision Support System).
2. Reduction in manual recording errors and expired product losses for test users.
3. Positive user feedback regarding credit tracking.
4. A project report and source code submitted for academic evaluation.

12. Conclusion

This project addresses a real and pressing problem faced by thousands of small retail businesses in Tanzania. By combining basic inventory management with intelligent decision support features such as expiry alerts, credit tracking, and purchase recommendations, the proposed system offers a practical, affordable, and locally relevant solution. The methodology, tools, and system requirements have been carefully defined to ensure successful implementation and validation.

13. References

(APA referencing style)

Mushi, R., & Mjema, E. (2021). Challenges of inventory management in small retail businesses in Tanzania. Journal of Business Research, 14(2), 45–59.

Tanzania Private Sector Foundation (TPSF). (2023). Digital readiness assessment for SMEs in Tanzania. Dar es Salaam: TPSF Publications.

National Bureau of Statistics Tanzania. (2022). Small and medium enterprises survey report. Dodoma: NBS.

Chen, Y., & Zhang, L. (2020). ERP systems for SMEs: A critical review. International Journal of Information Management, 51, 102–115.

Lema, J., & Mwita, S. (2022). Digital constraints for small retailers in East Africa. African Journal of Computing, 8(1), 33–47.

Kalinga, E. (2021). Adoption of ICT among small retailers in Tanzania. Tanzania Journal of ICT Research, 5(2), 22–38.

World Bank. (2021). Digital transformation of SMEs in Sub-Saharan Africa. Washington, DC: World Bank Group.

UNCTAD. (2022). Technology and innovation report 2022: Digital transformation for sustainable development. Geneva: United Nations.

14. Appendices

· Appendix A: Use case diagram (see Section 8.6.3.1).

· Appendix B: Class diagram (see Section 8.6.3.2).

· Appendix C: Data flow diagram Level 0 and Level 1 (see Section 8.6.3.3).